
```

%prob2
t_param = 1:5;
X_pts = [0, 1, 3, 2, 0];
Y_pts = [0, 2, 1, -1, 0];
% coeficientii Spline natural pentru x si y
[ax, bx, cx, dx] = spline_natural_coeffs(t_param, X_pts);
[ay, by, cy, dy] = spline_natural_coeffs(t_param, Y_pts);
t_fine = linspace(1, 5, 200);
X_spline = eval_spline(t_param, ax, bx, cx, dx, t_fine);
Y_spline = eval_spline(t_param, ay, by, cy, dy, t_fine);
figure('Name', 'Lab 08 - Problema 2: Spline Parametric');
plot(X_spline, Y_spline, 'm-', 'LineWidth', 2); hold on;
plot(X_pts, Y_pts, 'ko', 'MarkerFaceColor', 'k', 'MarkerSize', 6);
title('Problema 2: Curba Spline parametrica');
grid on; axis equal;

%prob3 - Litera "v"
x_letter = [1, 1.2, 2.0, 2.8, 3.0, 3.5];
y_letter = [4, 1.5, 0.0, 1.5, 4.0, 4.0];
n_pts = length(x_letter);
t_letter = 1:n_pts;
[aX, bX, cX, dX] = spline_natural_coeffs(t_letter, x_letter);
[aY, bY, cY, dY] = spline_natural_coeffs(t_letter, y_letter);
t_fine_letter = linspace(1, n_pts, 300);
X_eval_let = eval_spline(t_letter, aX, bX, cX, dX, t_fine_letter);
Y_eval_let = eval_spline(t_letter, aY, bY, cY, dY, t_fine_letter);
figure('Name', 'Lab 08 - Problema 3: Desenare Litera V');

%litera standard (v)
plot(X_eval_let, Y_eval_let, 'b-', 'LineWidth', 2); hold on;
plot(x_letter, y_letter, 'bo', 'MarkerFaceColor', 'b');
%litera de dimensiune dubla (v)
plot(2 * X_eval_let, 2 * Y_eval_let, 'r-', 'LineWidth', 2);
plot(2 * x_letter, 2 * y_letter, 'ro', 'MarkerFaceColor', 'r');
title('Problema 3: Litera Spline "v" Normala si Dublata (2x)');
legend('Litera 1x', 'Puncte 1x', 'Litera 2x', 'Puncte 2x', 'Location',
'best');
grid on; axis equal;

%animatie cu comet
disp('Apasa o tasta in Command Window pentru animatia literei v...');
pause;
figure('Name', 'Animatie Litera V');
comet(X_eval_let, Y_eval_let);
title('Problema 3(c): Animatie folosind comet (Litera v)');

%functie problema 1: calculul coeficientilor pentru spline
function [a, b_coeff, c, d] = spline_natural_coeffs(x_nodes, y_nodes)
    n = length(x_nodes) - 1;
    h = diff(x_nodes);
    a = y_nodes(:);

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alpha = zeros(n, 1);
for i = 2:n
    alpha(i) = 3/h(i)*(a(i+1) - a(i)) - 3/h(i-1)*(a(i) - a(i-1));
end

%algoritmul lui Thomas
l = ones(n+1, 1);
mu = zeros(n+1, 1);
z = zeros(n+1, 1);

for i = 2:n
    l(i) = 2*(x_nodes(i+1) - x_nodes(i-1)) - h(i-1)*mu(i-1);
    mu(i) = h(i) / l(i);
    z(i) = (alpha(i) - h(i-1)*z(i-1)) / l(i);
end

c = zeros(n+1, 1);
b_coeff = zeros(n, 1);
d = zeros(n, 1);

for j = n:-1:1
    c(j) = z(j) - mu(j)*c(j+1);
    b_coeff(j) = (a(j+1) - a(j))/h(j) - h(j)*(c(j+1) + 2*c(j))/3;
    d(j) = (c(j+1) - c(j))/(3*h(j));
end

a = a(1:n);
c = c(1:n);
end

% functie prob 1: evaluarea spline
function y_eval = eval_spline(x_nodes, a, b_coeff, c, d, x_eval)
    y_eval = zeros(size(x_eval));
    n = length(x_nodes) - 1;

    for k = 1:length(x_eval)
        x = x_eval(k);
        idx = find(x_nodes(1:end-1) <= x, 1, 'last');
        if isempty(idx), idx = 1; end
        if idx > n, idx = n; end
        dx = x - x_nodes(idx);
        y_eval(k) = a(idx) + b_coeff(idx)*dx + c(idx)*dx^2 + d(idx)*dx^3;
    end
end

%litera n
x_litera_N = [0, 0, 0.5, 1.0, 1.0, 1.0];
y_litera_N = [0, 4, 4.5, 4.5, 4, 0];

t_N = 1:length(x_litera_N);
[aX_N, bX_N, cX_N, dX_N] = spline_natural_coeffs(t_N, x_litera_N);

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[aY_N, bY_N, cY_N, dY_N] = spline_natural_coeffs(t_N, y_litera_N);
t_fine_N = linspace(1, length(t_N), 300);
X_eval_N = eval_spline(t_N, aX_N, bX_N, cX_N, dX_N, t_fine_N);
Y_eval_N = eval_spline(t_N, aY_N, bY_N, cY_N, dY_N, t_fine_N);

figure('Name', 'Litera n');
plot(X_eval_N, Y_eval_N, 'Color', '#D95319', 'LineWidth', 2.5); hold on;
plot(x_litera_N, y_litera_N, 'o', 'Color', '#FFBD33', 'MarkerFaceColor',
'#FFBD33', 'MarkerSize', 7);
title('Litera "n"');
grid on; axis equal; axis([-0.5 2 -0.5 5.5]);

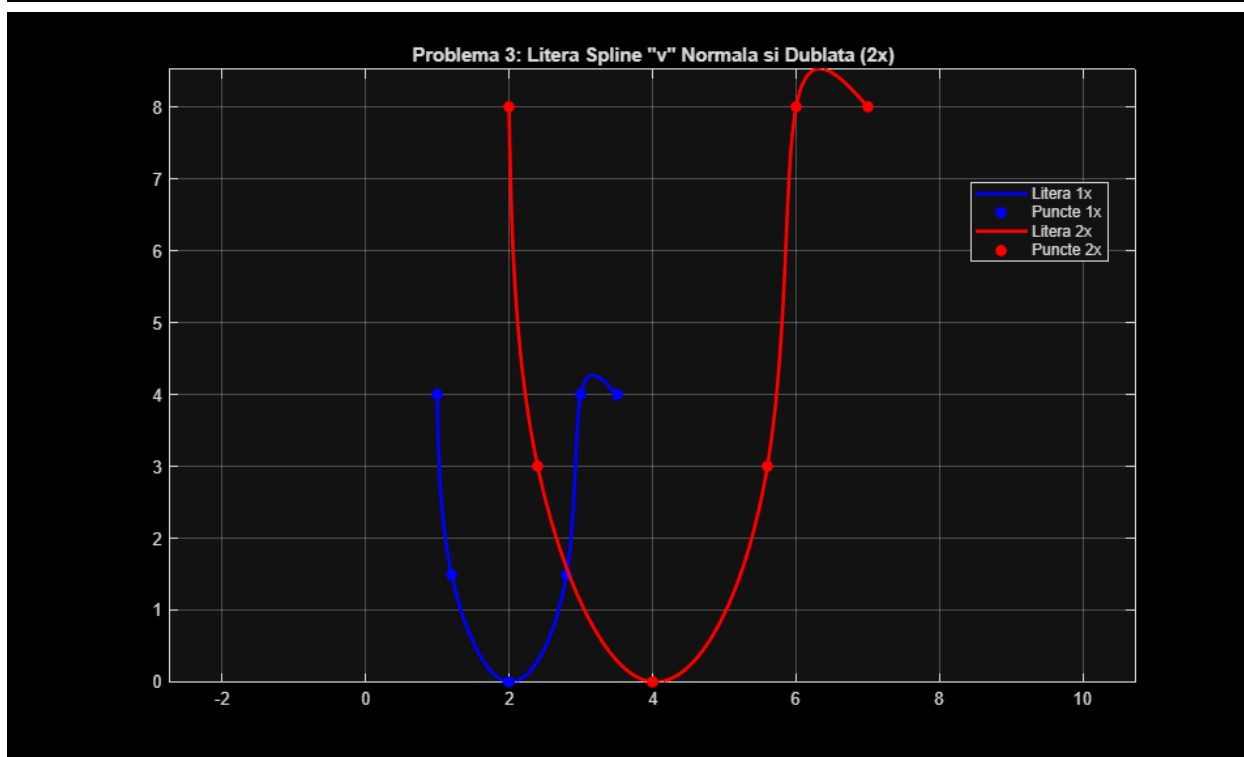
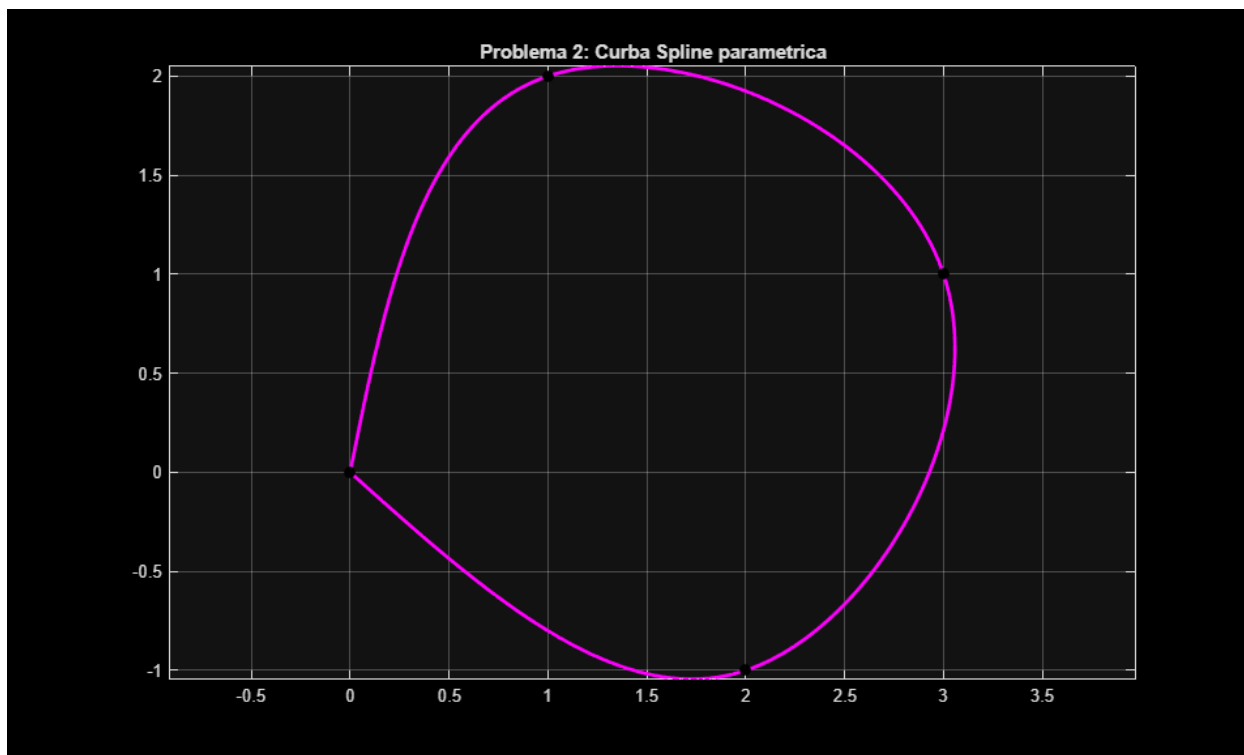
%litera m
x_litera_M = [0, 0, 0.5, 1.0, 1.0, 1.5, 2.0, 2.0, 2.0];
y_litera_M = [0, 4, 4.5, 4.5, 4, 4.5, 4.5, 4, 0];

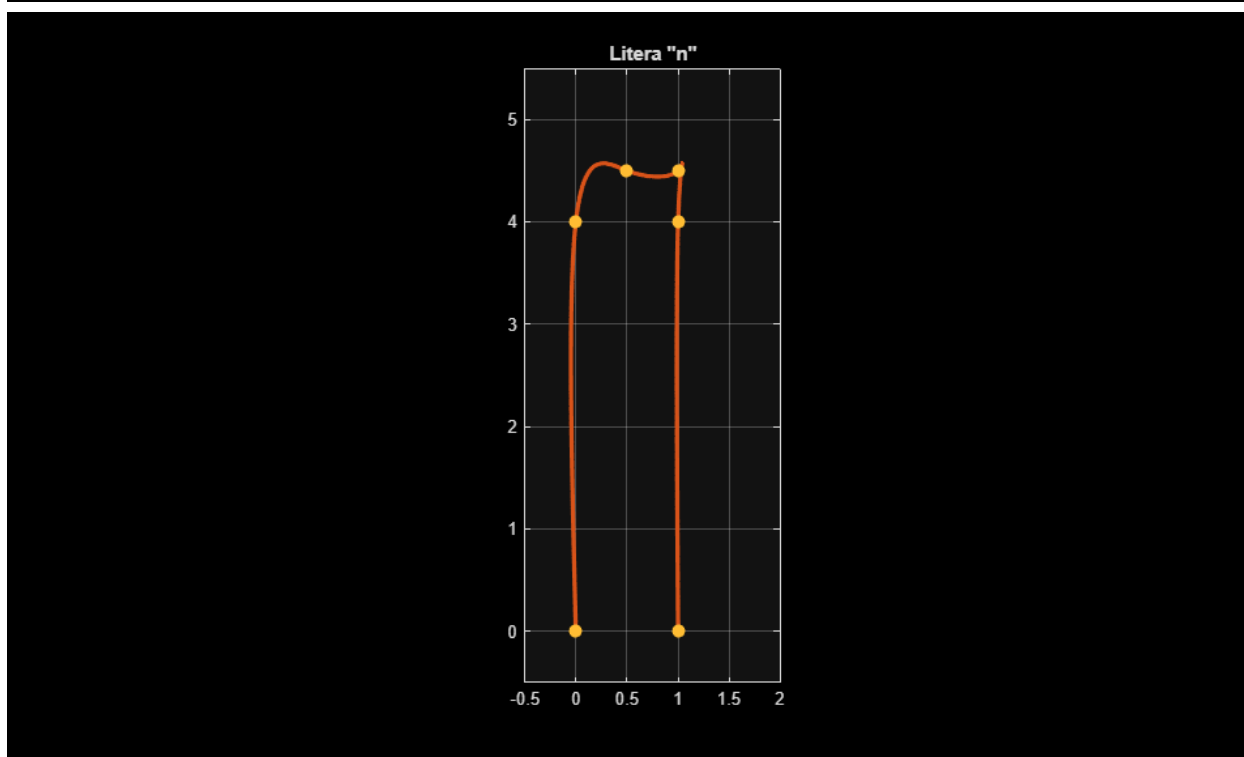
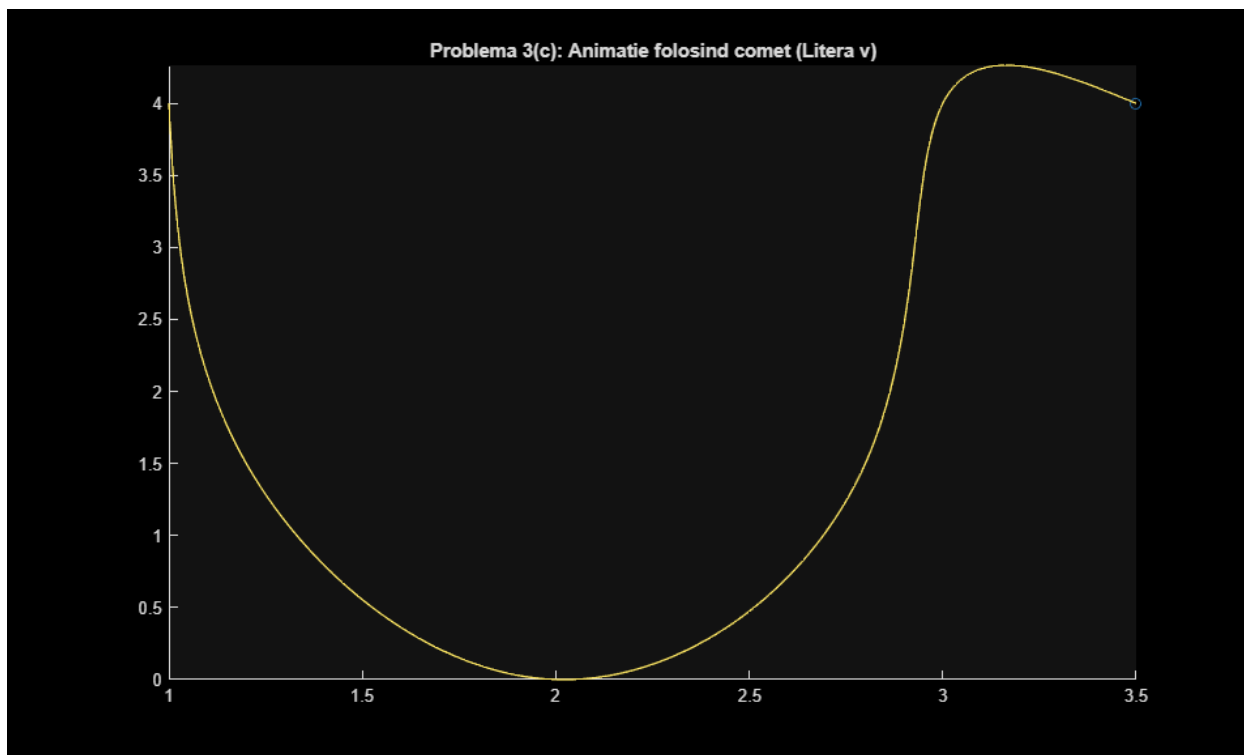
t_M = 1:length(x_litera_M);
[aX_M, bX_M, cX_M, dX_M] = spline_natural_coeffs(t_M, x_litera_M);
[aY_M, bY_M, cY_M, dY_M] = spline_natural_coeffs(t_M, y_litera_M);
t_fine_M = linspace(1, length(t_M), 400);
X_eval_M = eval_spline(t_M, aX_M, bX_M, cX_M, dX_M, t_fine_M);
Y_eval_M = eval_spline(t_M, aY_M, bY_M, cY_M, dY_M, t_fine_M);

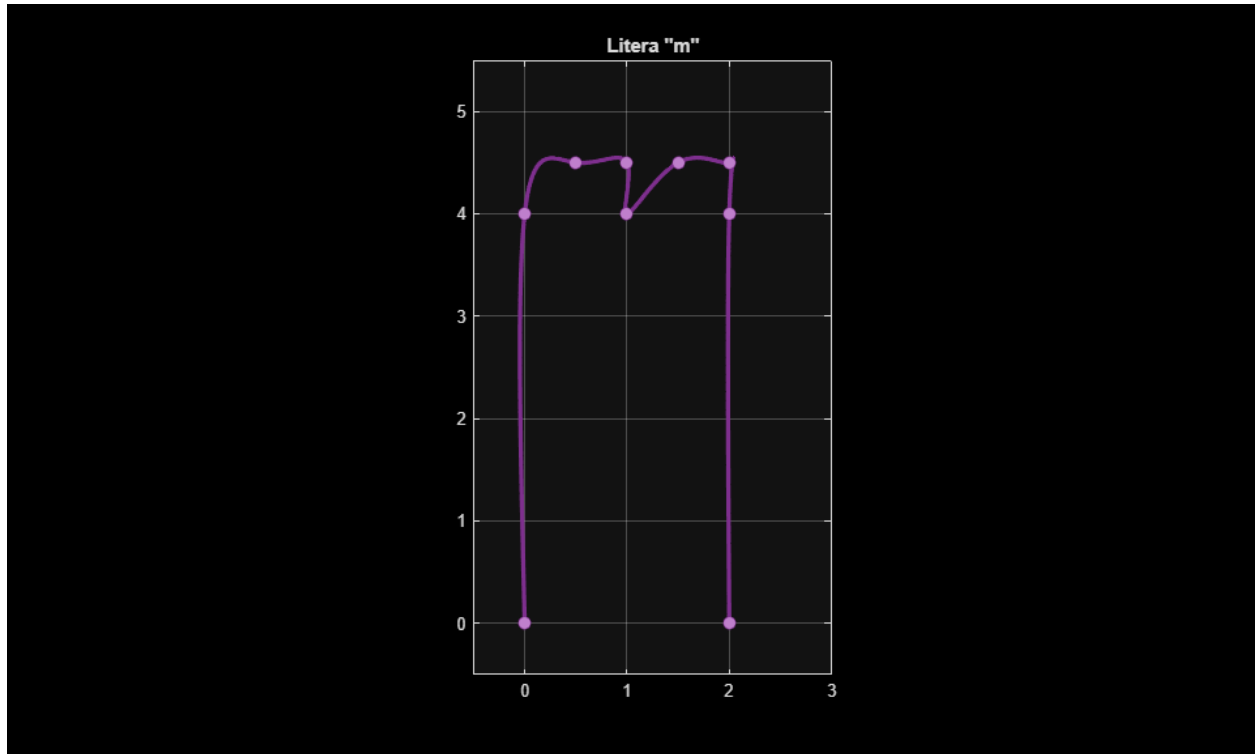
figure('Name', 'Litera m');
plot(X_eval_M, Y_eval_M, 'Color', '#7E2F8E', 'LineWidth', 2.5); hold on;
plot(x_litera_M, y_litera_M, 'o', 'Color', '#7E2F8E', 'MarkerFaceColor',
'#BF7FCC', 'MarkerSize', 7);
title('Litera "m"');
grid on; axis equal; axis([-0.5 3 -0.5 5.5]);

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Apasa o tasta in Command Window pentru animatia literei v...







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